



Moosic

Playlist for the soul



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Project Goal

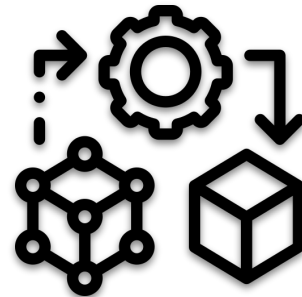


Problem: our experts are slow in creating new playlists!

**To create scalable and cohesive playlists using Spotify
audio features - initial prototype**



How We Built the Prototype



- We cleaned and prepared Spotify audio feature data
- We tested multiple scaling methods
- We used K-Means clustering
- We evaluated cluster cohesion manually and statistically

00:59



02:03





Feature Selection



danceability	energy	key
loudness	mode	speechiness
acousticness	instrumentalness	liveness
valence	tempo	time_signature

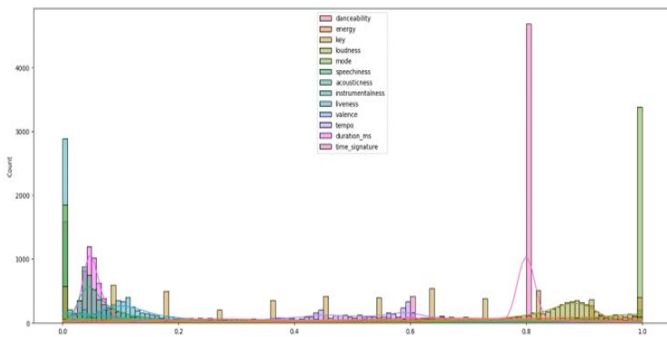
We removed:

- duration (as it does not define the song type and has extreme values)
- low-information variables (values consisting of strings)

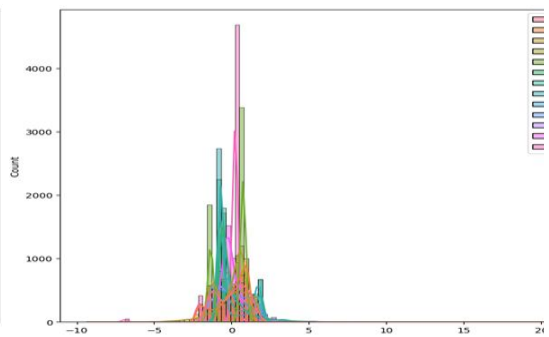
Scaling the Features



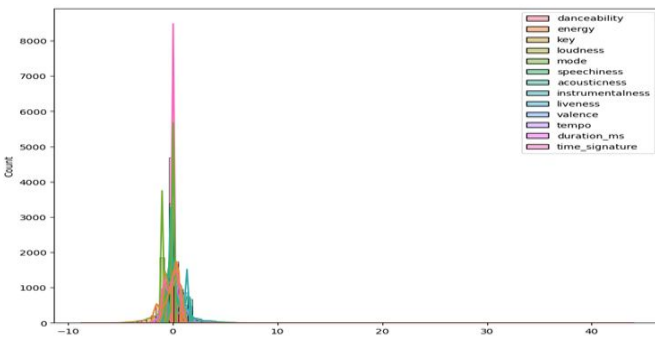
MinMax scaler



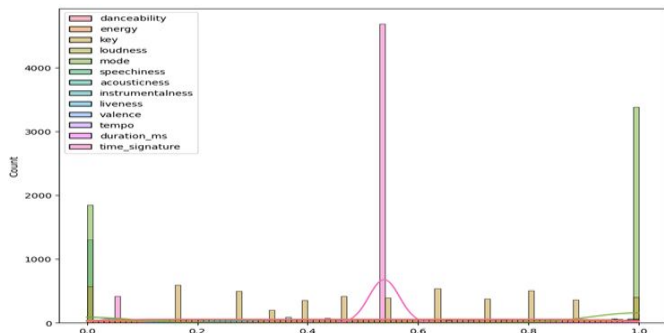
Standard scaler



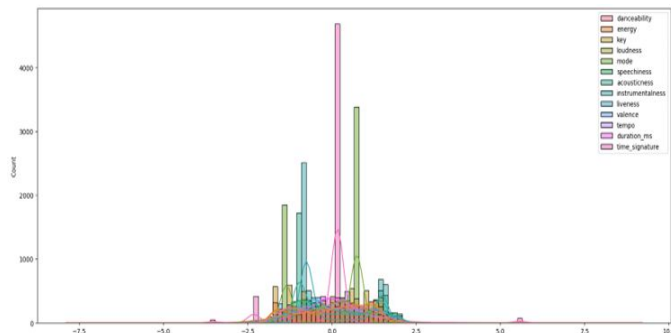
Robust scaler



Quantile Transformer scaler

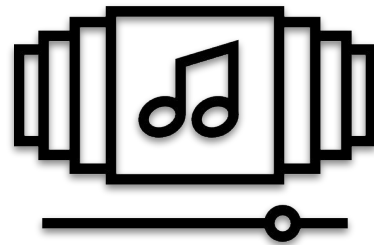


Power transformer scaler





How Many Playlists?



We evaluated multiple values of k using:

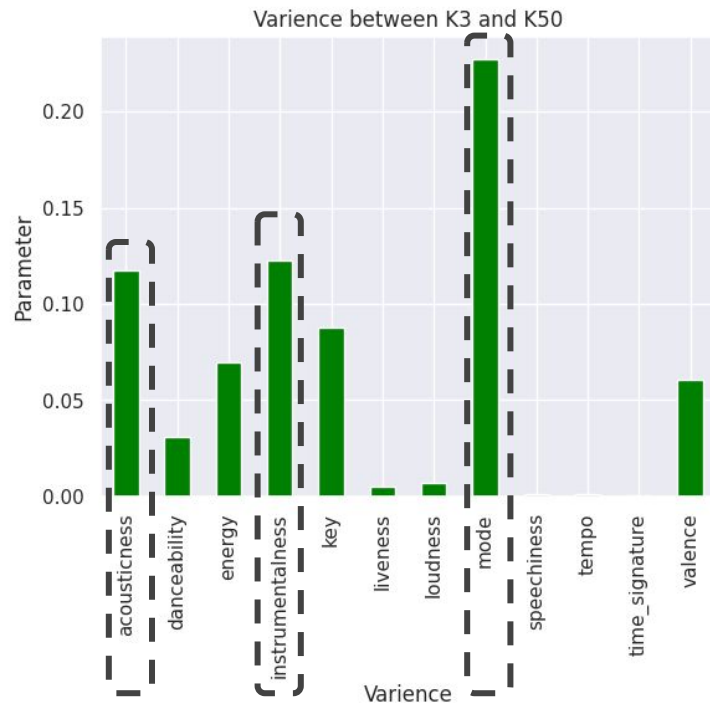
Elbow optimal k: 3

Silhouette optimal k: 2

Calinski optimal k: 50

Davies optimal k: 47

The final business prototype used **50** playlist clusters.





Prototype Evaluation



The generated playlists often grouped songs with:

- similar mood
- similar energy
- similar acoustic structure

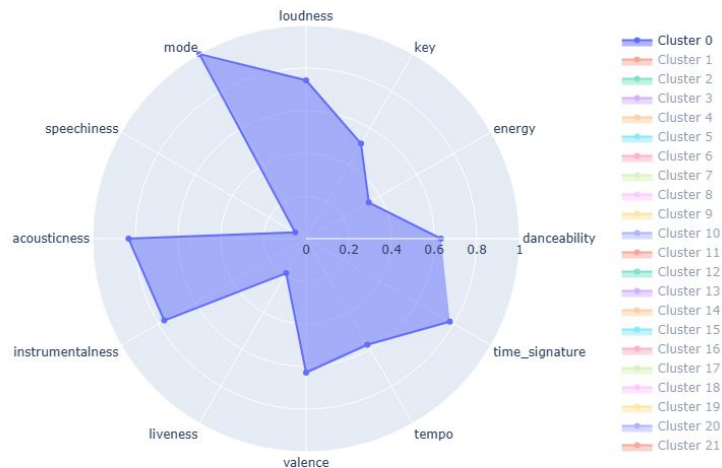
Some clusters still mixed genres despite similar audio profiles.



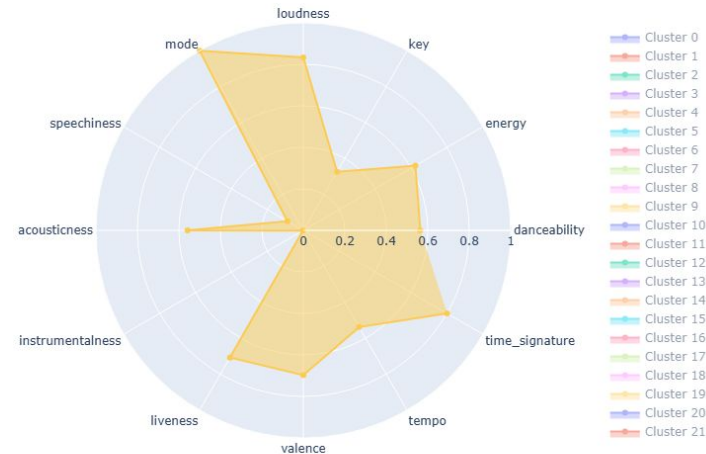
Example Playlist Cluster

We inspected smaller niche clusters: similar tempo,
similar emotional tone, Consistent energy levels
Randomly selected 3 clusters: **0, 44, 49**

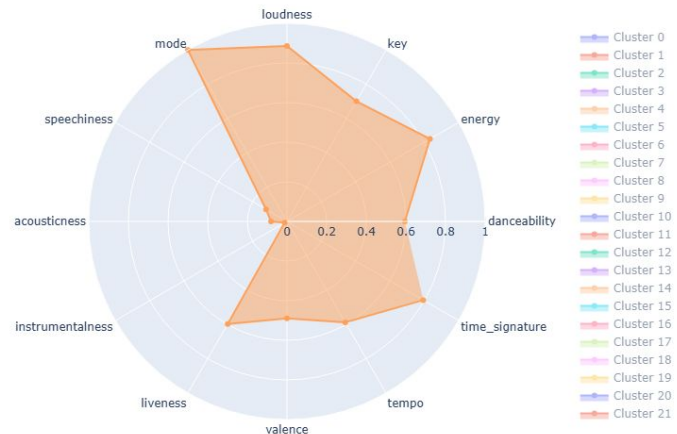
Radar chart of mean song character by cluster



Radar chart of mean song character by cluster



Radar chart of mean song character by cluster



Sample cluster



CLUSTER 38

Size: 257 songs

Interpretation: danceable, positive mood

Top feature means:

valence_standard: 1.09

danceability_standard: 0.99

mode_standard: 0.74

key_standard: 0.67

instrumentalness_standard: -0.65

music type detected: pop-rock / upbeat pop

	name	artist	cluster_calinski	music_style_calinski
103	Agua De Beber ...	Sophie Milman	38	pop-rock / upbeat pop
141	Magalenha ...	Sérgio Mendes	38	pop-rock / upbeat pop
166	Alegria ...	Batida	38	pop-rock / upbeat pop
168	Tirei o Chapéu ...	Batida	38	pop-rock / upbeat pop
206	I Wish You Were Here ...	Alpha Blondy	38	pop-rock / upbeat pop
...	...	...	...	...
5048	Picky ...	Joey Montana	38	pop-rock / upbeat pop
5049	Quiere Beber ...	Anuel AA	38	pop-rock / upbeat pop
5052	Con Altura ...	ROSALÍA	38	pop-rock / upbeat pop
5073	Hope ...	Jonathan Traylor	38	pop-rock / upbeat pop
5112	Increase My Faith ...	Brian Courtney Wilson	38	pop-rock / upbeat pop



Are Spotify Features Enough?



Spotify audio features capture measurable song characteristics well.

However, they do not fully capture:

- lyrics
- language
- cultural context
- trends
- artist relationships

ATTENTION!!!!

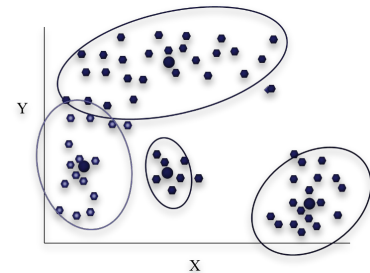
Spotify limited access to advanced audio-analysis endpoints because recommendation data became strategically valuable for AI and music recommendation systems.

Alternatives for obtaining audio features:

- existing public datasets (Kaggle, Million Song Dataset),
- Essentia audio analysis library,
- Librosa for extracting features directly from audio,
- Last.fm metadata,
- YouTube Music metadata,
- self-generated embeddings using deep learning models.



Is K-Means a Good Choice?



Pros

Cons

fast

Requires fixed number of cluster, eg. 50

scalable

sensitive to scaling

easy to explain

struggles with overlapping genres

provides a strong and scalable foundation for automated playlist generation

Some songs fit multiple music styles at the same time, but K-Means can assign each song to only one cluster

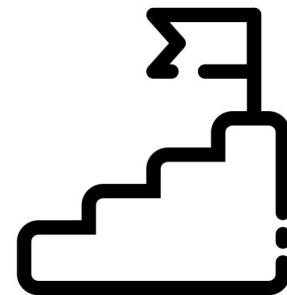
The prototype already produces cohesive playlists based on audio similarity.

To achieve production-level recommendation quality, the system should be expanded with user behavior data, embeddings, and more advanced clustering techniques.

The model captures acoustic similarity better than cultural or emotional similarity.



Next Steps



- additional test hierarchical clustering and DBSCAN
- add user behavior data
- include skip-rate and engagement metrics (users)
- evaluate playlists with real users

Thank you





Moosic Playlist Clustering Prototype

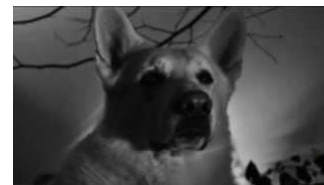


Machine Learning Playlist Generation Prototype: Sample music from sampled clusters

49



44



0

